

SORNA A

B.Tech – Electronics and Instrumentation Engineering
Puducherry Technological University, Puducherry
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CAREER OBJECTIVE

Electronics and Instrumentation Engineering graduate with technical knowledge and practical exposure to field instrumentation, industrial sensors and transducers, process measurement, control systems, and electronic circuits. Familiar with instrumentation applications, industrial automation, and process monitoring in industrial environments. Keen to strengthen core engineering expertise and develop practical skills in instrumentation, calibration, troubleshooting, and process control to contribute effectively to industrial operations and engineering projects.

EDUCATION

Puducherry Technological University, Puducherry

B.Tech – Electronics and Instrumentation Engineering | CGPA: 8.0

New Modern Vidhya Mandir

Higher Secondary Education | 82.33%

PROJECTS

Wearable Hand Tremor Monitoring System

- Developed a wearable tremor-monitoring concept using an MPU9250 9-axis IMU mounted on a wrist brace.
- Planned to analyze multi-axis motion data to characterize tremor patterns, with optional sEMG for muscle-activity information.
- Outlined a demo workflow combining motion filtering and adaptive actuator feedback for potential tremor compensation.

Drone-Inspired Environmental Monitoring System

- Developed a LabVIEW-based wireless environmental monitoring system using a Raspberry Pi Pico, DHT11, MQ-2, and resistive moisture sensor.
- Used an ESP8266 Wi-Fi link to send sensor readings for real-time monitoring.
- Planned the LabVIEW interface for live visualization, threshold alerts, and CSV data logging.

IntelliPump – Digital-Twin-Based Pump Health Monitoring

- Developed a machine-learning workflow using pump-related sensor readings to identify abnormal operating behaviour.
- Explored vibration, temperature, voltage, and current measurements for pump condition monitoring.
- Prepared a dashboard concept to visualize pump-health information and anomaly results; live sensor integration is in progress.

Vision-Based Navigation Using MASt3R + SLAM with Path Planning

- Developed a MATLAB-based UAV navigation simulation and explored MASt3R-SLAM and dense 3D reconstruction.
- Reviewed visual-odometry methods and A* path planning for autonomous navigation.

CERTIFICATIONS & WORKSHOPS

- Wildlife Ecology – NPTEL, 2024.
- Forest and Land Management – NPTEL, 2025.
- Biomedical Engineering Workshop – NIT Puducherry, 2023.
- IoT Made Easy with ESP32 – VIT Vellore, March 2026.
- Design Thinking Workshop – Electronics and Instrumentation (E&I;) Department, Puducherry Technological University (PTU).

INTERNSHIP EXPERIENCE

Research Intern – NITTTR Chennai | June 2026

- Worked on “Vision-Based Navigation Using MASt3R + SLAM with Path Planning”.
- Performed MATLAB simulation and explored dense visual SLAM techniques.

Intern – Thermal Power Plant, NLC India Limited, Neyveli

- Studied boilers, turbines, generators, auxiliary equipment, and industrial instrumentation systems.
- Gained knowledge of lignite-based power generation and safety procedures.

Intern – Cornerstone Engineering Solutions, Puducherry

- Learned fundamentals of programmable logic controllers (PLCs), industrial relays, and basic SCADA.
- Gained hands-on exposure to relay and PLC operation and introductory industrial automation practices.

TECHNICAL SKILLS

Electronics Skills: Circuit design using diodes, transistors, FETs, SCR, UJT, DIAC, TRIAC, and operational amplifiers (Op-Amps); knowledge of amplifiers, oscillators, rectifiers, multivibrators, and wave-shaping circuits.

Instrumentation & Process Control: Sensors and transducers, temperature and pressure measurement, level and flow measurement, industrial instrumentation, process control, control valves, and PID controllers.

Embedded Systems & Hardware: Arduino UNO, LPC2148 ARM, 8051 microcontroller, 8085 microprocessor, and sensor interfacing (eye blink and IR sensors).

PLC & Industrial Automation: PLC fundamentals, analog and digital I/O, 4–20 mA signal basics, and SCADA fundamentals.

Software & Simulation Tools: Arduino IDE, Proteus, LabVIEW, Keil µVision, MATLAB, and Simulink.

ACHIEVEMENTS

- Second Prize – Mini Project Poster Presentation for a Fruit Ripeness Sensor project, organized by the Electronics and Instrumentation (E&I;) Department, Puducherry Technological University (PTU).
- Second Prize – Quiz Competition: Jigyasa 2026, organized by the Electronics and Instrumentation (E&I;) Department, Puducherry Technological University.
- Second Prize – Coco: Sports competition organized by the Sports Department, Puducherry Technological University.